

# Tema 8

## Non-Normal Distributions of Returns

Many portfolio-based finance theories utilize a mean-variance framework in which investors make decisions on the basis of means and variances of the rates

of return. For example, the CAPM assumes all asset returns are normally distributed or investors have mean-variance preferences. Portfolio performance measures such as the Sharpe measure, the Treynor measure, and the Jensen measure are based on the two parameters of the normal distribution. However, there has been empirical evidence that asset returns are not normally distributed since Mandelbrot (1963), Fama (1965), and Blattberg and Gonedes (1974).<sup>1</sup> Researchers observe nonsymmetric, highly peaked, and longer-tailed (leptokurtic) characteristics in the empirical unconditional distribution of asset returns. Table 1 shows statistical results from normality tests of international stock market return distributions from 14 developed countries. Most of the countries show significantly positive skewness. In particular, skewness and kurtosis coefficients using weekly returns and monthly returns are quite different, even though skewness and kurtosis coefficients are pure index numbers that should remain invariant to the return measurement interval.<sup>2</sup> This evidence might be indicating that the returns are generated from several nonidentical distributions, such as a mixture of normal distributions.

In a dynamic economic environment, firms' investment and financing decisions will affect the systematic risk, expected return, and standard deviation of returns. For example, Boness, Chen and Jatusipitak (1974) find parameters shift after a capital structure change and Christie (1982) demonstrates that the standard deviation of a stock's return is an increasing function of both financial and operating leverage. Beaver's (1968) realized returns and Patell and Wolfson's (1981) *ex ante* assessments support an increase in the variance of stock returns around the announcement of quarterly earnings. Macro information shocks may also shift the level of interest rates and market risk premiums. Therefore, it is not surprising for the expected return and the risk dimensions of individual stocks and portfolios to change through time. Nonstationarity of mean and risk could cause the skewness, highly peaked, and longer-tailed (leptokurtic) characteristics in the empirical unconditional distribution of asset returns. This chapter presents several statistical distribution models that can capture the skewness and leptokurtic features of asset returns.

### **1 STABLE PARETIAN DISTRIBUTIONS**

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This distribution is defined by its characteristic function because its probability density is generally not known. Stable Paretian distributions are completely characterized by four parameters. The first parameter, denoted  $\alpha \in (0, 2)$ , is known as the

**TABLE 1** Summary Statistics of International Stock Market Return Distributions

|             | Mean return |        | Variance |       | Skewness <sup>a</sup> |        | Kurtosis <sup>a</sup> |         | W-statistic |       | Prob < W <sup>b</sup> |       |
|-------------|-------------|--------|----------|-------|-----------------------|--------|-----------------------|---------|-------------|-------|-----------------------|-------|
|             | Week        | Month  | Week     | Month | Week                  | Month  | Week                  | Month   | Week        | Month | Week                  | Month |
| Australia   | 0.176       | 0.767  | 0.152    | 3.021 | 0.199                 | 28.799 | 3.351                 | 531.676 | 0.986       | 0.957 | 0.75                  | 0.05  |
| Belgium     | 0.218       | 0.893  | 0.164    | 2.815 | 0.504                 | 28.591 | 5.437                 | 419.759 | 0.983       | 0.937 | 0.42                  | 0.00  |
| Canada      | 0.075       | 0.342  | 0.121    | 0.756 | 0.390                 | 1.133  | 2.032                 | 19.377  | 0.982       | 0.955 | 0.29                  | 0.03  |
| France      | 0.278       | 1.170  | 0.165    | 3.906 | -0.017                | 42.870 | 3.083                 | 889.654 | 0.986       | 0.958 | 0.69                  | 0.05  |
| Germany     | 0.248       | 1.052  | 0.159    | 2.153 | -0.212                | 18.717 | 2.489                 | 287.944 | 0.982       | 0.948 | 0.33                  | 0.01  |
| Hong Kong   | 0.550       | 2.210  | 0.212    | 1.140 | 0.354                 | 2.368  | 4.647                 | 65.767  | 0.975       | 0.955 | 0.03                  | 0.03  |
| Italy       | 0.044       | -0.027 | 0.145    | 2.672 | 0.102                 | -3.563 | 3.179                 | 268.171 | 0.974       | 0.944 | 0.01                  | 0.01  |
| Japan       | 0.042       | 0.250  | 0.126    | 2.429 | 0.241                 | 9.454  | 4.091                 | 212.333 | 0.973       | 0.932 | 0.01                  | 0.00  |
| Netherlands | 0.243       | 1.024  | 0.052    | 0.926 | 0.021                 | -0.917 | 0.291                 | 25.572  | 0.972       | 0.973 | 0.00                  | 0.33  |
| Singapore   | 0.409       | 1.607  | 0.323    | 0.934 | 1.052                 | 2.006  | 14.101                | 26.943  | 0.960       | 0.951 | 0.00                  | 0.02  |
| Sweden      | 0.251       | 1.007  | 0.184    | 1.886 | 0.222                 | 7.176  | 3.426                 | 114.606 | 0.988       | 0.971 | 0.85                  | 0.28  |
| Switzerland | 0.303       | 1.206  | 0.100    | 1.703 | -0.163                | -0.744 | 1.694                 | 90.204  | 0.984       | 0.948 | 0.55                  | 0.01  |
| U.K.        | 0.170       | 0.692  | 0.079    | 1.083 | 0.035                 | 0.816  | 1.125                 | 47.455  | 0.979       | 0.970 | 0.15                  | 0.25  |
| U.S.        | 0.220       | 0.943  | 0.132    | 0.429 | 0.427                 | 0.368  | 2.323                 | 9.565   | 0.981       | 0.952 | 0.25                  | 0.02  |

<sup>a</sup>Skewness and kurtosis represent the third and fourth central moments of the rates of return, respectively.

<sup>b</sup>If the probability (prob < W) is less than 0.10, the null hypothesis of normality cannot be supported at the ten percent level of significance.

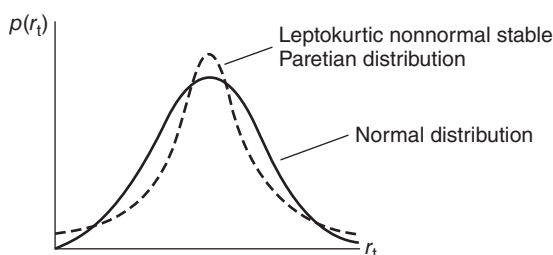
Source: Adapted from Chunchachinda, Krishnan, Hamid, and Prakash, "Portfolio Selection and Skewness: Evidence from International Stock Markets," *Journal of Banking and Finance* 21 (1997): 143–167.

characteristic exponent and determines the membership of the particular distribution. For example, if  $\alpha = 2$ , then the random variable has a normal distribution, and if  $\alpha = 1$ , then the random variable has a Cauchy distribution. The second, third, and fourth parameters determine the location, dispersion, and symmetry of the random variable's distribution and, for a normal distribution, correspond to mean, standard deviation, and skewness (which, for the normal, is zero), respectively.

The normal distribution is one particular member of what is known as the family of stable Paretian distributions. The common characteristic of all members of this family is that they are invariant under addition, meaning that when random variables from a particular distribution are added together, the resulting random variable will have the same distribution. For example, if the returns on firm A and firm B are from Cauchy distributions (a particular membership of the stable Paretian distribution family), then the returns on any portfolio formed by combining A and B will also conform to the characteristics of the Cauchy distribution.

The reasonableness of invoking the central limit theorem to argue that returns are normally distributed can be questioned by noting that this theorem is applicable only when the distribution from which the sample is drawn has a finite standard deviation.<sup>3</sup> In particular, if daily returns come from a non-normal stable Paretian distribution (meaning  $\alpha \neq 2$ ), then the central limit theorem cannot be invoked and the conclusion that monthly returns will be normally distributed cannot be drawn. Indeed, it has been argued conceptually that the monthly distribution for common stock returns is leptokurtic, meaning the distribution has more observations in the tails and center and fewer in the shoulders relative to the normal distribution. An illustration of such a distribution is shown in Figure 1. The reasoning behind this argument is that returns are affected by information, and information is generally in one of two categories—either large importance or small importance. Such a dichotomous information structure tends to generate either large or small returns. Essentially, returns are leptokurtic because information is leptokurtic.

Fama (1965), in a detailed empirical study of daily returns, found that observed returns were leptokurtic and concluded that a non-normal stable Paretian distribution fit the data better than a normal distribution.<sup>4</sup> An implication of his study is that the distribution for common stock returns has no finite standard deviation, suggesting that the central limit theorem–based argument for normality is inappropriate. Another implication is that even if investors have quadratic utility functions, they still cannot make decisions on the basis of means and standard deviations



**FIGURE 1** Normal and Leptokurtic Non-normal Stable Paretian Distributions

because standard deviations do not exist. However, by redefining the measure of risk, two-dimensional analysis of portfolios can still be performed. In this situation what is known as the semi-interquartile range can be used as the risk measure.<sup>5 6</sup>

## 2 THE STUDENT'S $t$ -DISTRIBUTION

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The Student's  $t$  density function of asset returns,  $r_t$ , with location parameter  $\mu$ , scale parameter  $H$ , and degrees of freedom  $\nu$  is represented by

$$f(r_t|\Theta) = \frac{\Gamma((\nu+1)/2)\sqrt{H}}{\Gamma(\nu/2)\sqrt{\pi\nu}} \left[1 + \frac{H}{\nu}(r_t - \mu)^2\right]^{-(\nu+1)/2} \quad (1)$$

where  $\Theta = \{\mu, H, \nu; -\infty < \mu < \infty, H > 0, \nu > 0\}$ ,  $\mu$  is the mean value of  $r_t$ , and  $\Gamma(\cdot)$  is the gamma function. The Student's- $t$  distribution has the following properties: (1)  $E(x) = \mu$  for  $\nu > 1$ , and undefined otherwise, and  $\text{Var}(x) = H^{-1}[\nu/(\nu-2)]$  for  $\nu > 2$ , and  $\infty$  for  $1 < \nu \leq 2$ , (2) all moments of order  $k < \nu$  are generally finite, (3) when  $\nu = 1$ , the Student's- $t$  distribution is the Cauchy density function, and (4) as the degrees of freedom  $\nu \rightarrow \infty$ , the Student's- $t$  distribution converges to the normal distribution. Because the Student's- $t$  distribution is symmetric, the mean, median, and mode are the same. The skewness is zero for  $\nu > 3$ , and the kurtosis is  $6/(\nu-4)$  for  $\nu > 4$ .

Blattberg and Gonedes (1974) argued that the observed leptokurtic daily return distribution can be explained by an alternative to the non-normal stable Paretian distribution, the Student's- $t$  distribution. Although able to explain the leptokurtosis observed by Fama (1965) in daily returns, the Student's- $t$  distribution has a finite standard deviation. Thus, the central limit theorem can be applied to it and used to provide an argument that monthly returns are approximately normal. Furthermore, Blattberg and Gonedes, and later Fama (1976), argue that the empirical evidence on monthly returns suggests that normality is a reasonable approximation.

## 3 MIXTURES OF NORMAL DISTRIBUTIONS

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In this distribution model, asset returns are normally distributed, but there is more than one such distribution per stock, meaning that the return distribution shifts through time and, thus, is not stationary.

### 3.1 Discrete Mixtures of Normal Distributions

Kon (1984) pointed out that the arguments for the Student's- $t$  distribution or the stable Paretian distribution generally assume that asset returns are from the same distribution. Because there are good reasons to expect this distribution to be shifting or nonstationary, Kon investigated whether or not daily returns could be modeled as a discrete mixture of different normal distributions in which each return observation is viewed as a drawing from one of a finite number of normal distributions with some probability. The observations are not necessarily consecutive. He found they could be so modeled and, in a comparison with the Student's- $t$  distribution, found the mixed distribution model to have more descriptive validity.

This view suggests that each return observation on a stock,  $r_t$ , has been generated by one of the following  $L$  distinct equations (or regimes):

$$\begin{aligned} r_t &= \mu_1 + u_{1t} & t \in I_1 \\ r_t &= \mu_2 + u_{2t} & t \in I_2 \\ &\dots \\ r_t &= \mu_L + u_{Lt} & t \in I_L \end{aligned}$$

where  $\mu_j$  is the mean of the return,  $r_t$ , in the  $j$ -regime ( $j = 1, 2, \dots, L$ ),  $I_j$  is the homogeneous information set with  $T_j$  observations in each set.  $\sum_{j=1}^L T_j = T$ ,  $u_{jt}$  is independently and identically normally distributed with a mean of zero and variance of  $\sigma_j^2$ ,  $0 < \sigma_j^2 < \infty$ ,  $j = 1, 2, \dots, L$ . Define  $\lambda_j = T_j/T$  as the proportion of observations associated with information set  $I_j$ . Thus, the probability density function for  $L$  regimes is described as

$$f(r_t | \Theta) = \sum_{j=1}^L \lambda_j \frac{1}{\sqrt{2\pi\sigma_j^2}} \exp \left\{ -\frac{1}{2\sigma_j^2} (r_t - \mu_j)^2 \right\} \quad (2)$$

where  $\lambda_j$  are the mixing probabilities, and the parameter vector is  $\Theta = \{\mu_j, \sigma_j^2, \lambda_j\}$  for  $j = 1, \dots, L$ .

### 3.2 Sequential Mixtures of Normal Distributions

Kim and Kon (1996, 1999) argued that relevant information that changes the risk-return structure is randomly released within some time interval, not at every moment, but in sequence. These information events translate into *sequential* discrete structural shifts for the mean and/or variance parameters in the time-series of security returns. To accommodate this sequential shift, they suggest a sequential mixture of normal distribution models in which asset returns are a sequence of stock returns that are normal variates with stationary mean or variance parameters up to a shift point. Define these as  $\tau_1$  return observations from regime 1. Thereafter, returns  $\tau_1 + 1$  to  $\tau_2$  are a sequence of return observations from regime 2, and so forth, until the last sequence of return observations from  $\tau_{K-1} + 1$  to  $\tau_K$  is the  $K$ -th regime. The formulation of this model is as follows:

$$\begin{aligned} \text{Regime 1:} \quad r_t &= \mu_1 + \varepsilon_{1t}, & t &= 1, \dots, \tau_1 \\ \text{Regime 2:} \quad r_t &= \mu_2 + \varepsilon_{2t}, & t &= \tau_1 + 1, \dots, \tau_2 \\ &\vdots \\ \text{Regime K:} \quad r_t &= \mu_K + \varepsilon_{Kt}, & t &= \tau_{K-1} + 1, \dots, \tau_K (= T) \end{aligned} \quad (3)$$

where  $\mu_k$  is the mean of the return,  $r_t$ , in the  $k$ -regime ( $k = 1, 2, \dots, K$ ) and  $\varepsilon_{kt}$  is the random error term of the  $k$ -th regime with mean 0 and variance  $\sigma_{\varepsilon_k}^2$

( $k = 1, \dots, K$ ). The nonstationarity arises from observations that are drawn from a different population after each shift point.

The parameter vector to be estimated is  $\Theta = \{\mu_k, \sigma_k^2, \tau_k\}$  for  $k = 1, \dots, K$ . However, the key parameter to be estimated is  $\tau_k$ , because  $\mu_k$  and  $\sigma_k^2$  can be estimated with the sample mean and variance after detecting the shift points,  $\tau_k$ . The estimated shift points reflect announcement of information events such as merger and acquisition and information leakage. The estimation procedure is described in Kim (1991) and Kim and Kon (1996, 1999). They found that the sequential mixture specification has substantially more descriptive ability than the competing alternatives (normal distribution, Student's- $t$  distribution, discrete mixture of normal distributions, and Poisson jump-diffusion process model).

## 4 POISSON JUMP-DIFFUSION PROCESS

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Press (1968), Jarrow and Rosenfeld (1984), Ball and Torous (1985), and Akgiray and Booth (1986, 1987) provide some positive evidence on a Poisson jump-diffusion process for the distribution of stock returns. The Poisson jump-diffusion process for the distribution of stock returns consists of a geometric Brownian motion component and an independent compound Poisson process with normally distributed jump amplitudes. This Poisson jump-diffusion process can be expressed as follows:

$$\log \left( \frac{P_t}{P_{t-1}} \right) = \left( \alpha - \frac{1}{2} \sigma^2 \right) t + \sigma Z_t + \sum_{n=1}^{N_t} J_n \quad (3)$$

where  $P_t$  is the security price at time  $t$ ,  $Z_t$  is a standard Brownian motion,  $N_t$  is a Poisson counting process with parameter  $\lambda > 0$ ,  $J_n$  is a normal random variate with mean  $\mu_J$  and variance  $\sigma_J^2$  representing the logarithm of one plus the percentage change in security price caused by the  $n$ -th jump,  $\alpha$  is the instantaneous conditional expected rate of return per unit of time for the Brownian motion part of the process, and  $\sigma^2$  is the instantaneous conditional variance. The resulting probability density function of a security return is

$$f(r_t | \Theta) = \sum_{n=1}^{\infty} \frac{e^{-\lambda} \lambda^n}{n!} \phi(r_t | \mu + n\mu_J, \sigma^2 + n\sigma_J^2) \quad (4)$$

where  $\mu = \alpha - (1/2) \sigma^2$ ,  $\Theta = \{\mu, \sigma^2, \lambda, \mu_J, \sigma_J^2\}$ , and  $\phi(a|b)$  denotes a normal density function with mean  $a$  and variance  $b$ . This model is essentially an infinite mixture of normal distributions with parameter restrictions.

## 5 LOGNORMAL DISTRIBUTIONS

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Logarithmic properties are applied in many ways in finance. Logarithmic utility and using natural logarithm to compute continuously compounded returns are useful applications for logarithms. Lognormal probability distributions are another beneficial application.

## 5.1 Specifications of Lognormal Distributions

One argument against normality is based on the concept of the investor's limited liability. There is a positive probability of achieving a return below any given value if returns are assumed to be normal. However, limited liability law means investors cannot lose more than the value of their investment. In other words, liability means that investors cannot earn less than  $-100$  percent. Thus, on conceptual grounds it can be seen that normality and limited liability are inconsistent. Furthermore, although limited liability restricts how low returns can be, there are no limits on how high returns can be. That is, gains yielding rates of return in excess of  $100$  percent are not uncommon. This suggests that returns are likely to be generated by a positively skewed distribution.<sup>7</sup> As a result of limited liability and positive skewness, a lognormal distribution is often assumed to be more appropriate than a normal distribution in describing security returns.<sup>8</sup> Even though this skewness diminishes with the length of the differencing interval (that is, the holding period), positive skewness is still found in returns measured over periods as short as a month for both individual stocks and mutual funds.

When a probability distribution or a frequency distribution is positively skewed, its mean is larger than its median, and its median is larger than its mode. But it is not known which of these three measures of central tendency investors use in forming their expectations of the central or most likely rate of return. As a result, both the mean and variance statistics on which portfolio theory is based may be biased. To see this possible bias, let  $M$  denote the mode (or median) rate of return. If investors perceive  $M$  to be the central or most likely rate of return, so that they base their investment decisions on  $M$  instead of the mean,  $\mu$ , then the difference  $[\mu - M]$  measures bias. This bias affects the mean-squared error (MSE):

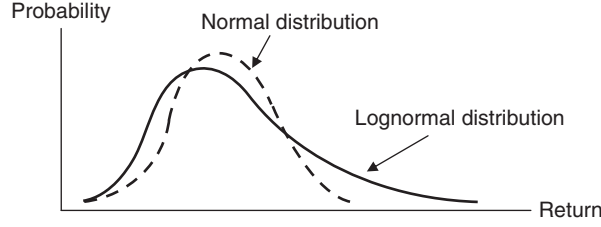
$$\begin{aligned} \text{MSE}(r) &= E(r - M)^2 \\ &= E\{[r - E(r)] - [M - E(r)]\}^2 \\ &= E\{[r - E(r)]^2 + [M - E(r)]^2\} \\ &= \text{Var}(r) + \text{bias}^2 \end{aligned} \tag{5}$$

Equation (5) shows that the second statistical moment,  $\text{Var}(r)$ , differs from the mean-squared errors around  $M$ . It is possible that skewness causes such bias by causing investors to focus on  $M$  instead of  $E(r)$ .<sup>9</sup>

It also can be shown that even if  $\mu$  is the measure of central tendency used by investors, skewness will nevertheless cause the alpha and beta statistics in the single-index market model, to be biased and inconsistent. The problem is that portfolio theory and ordinary least-squares (OLS) regression both consider only the first two statistical moments. As result, skewness causes biases that depreciate the value of portfolio theory and OLS regression statistics.

Skewness can be reduced and even eliminated from many probability distributions by transforming the (positively skewed) holding period return to the continuously compounded return.

$$\dot{r}_t = \ln\left(\frac{P_t}{P_{t-1}}\right) = \ln(1 + r_t) \tag{6}$$



**FIGURE 2** Normal and Lognormal Distributions

where  $\dot{r}_t$  is the continuously compounded return and  $r_t$  is the holding period return.<sup>10</sup> If the continuously compounded return,  $\dot{r}_t$ , has a normal distribution with mean  $\dot{\mu}$  and variance  $\dot{\sigma}^2$  so that  $\dot{r}_t \sim N(\dot{\mu}, \dot{\sigma}^2)$ , then the holding period return (or discrete period return),  $1 + r_t$  (or simply  $r_t$ ), is said to have a *lognormal* distribution. Then, the mean and variance of  $1 + r_t$  are denoted

$$E(1 + r_t) = \mu = \exp\left(\dot{\mu} + \frac{1}{2}\dot{\sigma}^2\right) \quad (7)$$

$$\begin{aligned} \text{Var}(1 + r_t) &= \sigma^2 = \exp(2\dot{\mu} + 2\dot{\sigma}^2) - \exp(2\dot{\mu} + \dot{\sigma}^2) \\ &= \mu^2 [\exp(\dot{\sigma}^2) - 1] \end{aligned} \quad (8)$$

The median of  $1 + r_t$  is  $\exp(\dot{\mu})$ , and the mode of  $1 + r_t$  is  $\exp(\dot{\mu} - \dot{\sigma}^2)$ . In positively skewed distributions such as lognormal distributions, mean < median < mode. Figure 2 illustrates the lognormal and normal probability distributions of  $r_t$  and  $\dot{r}_t$ , respectively.

As shown in Figure 2, the logarithmic transformation “pulls in” the skewed positive tail of the distribution of asset returns and extends its truncated negative tail to minus infinity to create a more symmetrical (and hopefully more nearly normal) distribution. It is appropriate to use the lognormal probability distribution when the natural logarithm of returns has a normal distribution.

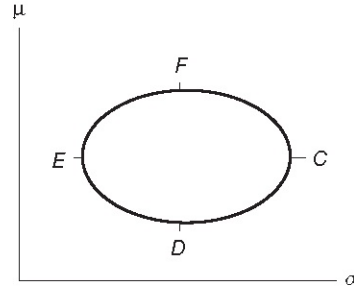
Because  $\dot{r}_t$  is normally distributed, its measure of skewness (the third moment around its mean, equal to  $E[(\dot{r}_t - \dot{\mu})^3]$ ) will be equal to zero. However, if  $r_t$  (or  $1 + r_t$ ) is lognormally distributed, it will have a positive third moment. This moment, denoted  $\kappa$ , will be functionally related to  $\dot{\mu}$  and  $\dot{\sigma}^2$  as follows:

$$\kappa = \dot{\mu}^3 \left[ \left( \frac{\dot{\sigma}^2}{\dot{\mu}^2} \right)^3 + 3 \left( \frac{\dot{\sigma}^2}{\dot{\mu}^2} \right)^2 \right] \quad (9)$$

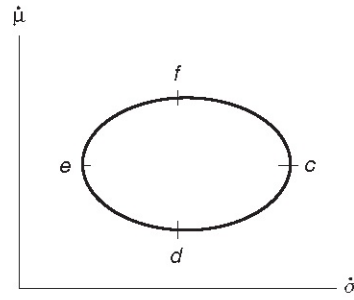
More generally, higher moments of  $r_t$  (or  $1 + r_t$ ) will be functions of  $\dot{\mu}$  and  $\dot{\sigma}$ , and thus the only parameters necessary to describe  $r_t$  are  $\dot{\mu}$  and  $\dot{\sigma}$ . As a result, while being slightly reformulated, portfolio theory still involves only means and standard deviations when returns are lognormally distributed.

## 5.2 Portfolio Analysis under Lognormality

Assuming that returns are lognormally distributed does not negate the use of mean-variance analysis. If the assumption of normality is replaced by an assumption of



**FIGURE 3A** Opportunity Sets in Risk-return Space  $(\mu, \sigma)$  for Noncompounded Returns



**FIGURE 3B** Opportunity Sets in Risk-return Space  $(\dot{\mu}, \dot{\sigma})$  for Continuously Compounded Returns

lognormality and investors are assumed to have utility functions displaying positive marginal utility of wealth, then a modified form of the efficient frontier theorem is inappropriate. Figures 3A and 3B display ellipses within which the opportunity set of risky securities lies in  $(\mu, \sigma)$  space and  $(\dot{\mu}, \dot{\sigma})$  space, respectively.<sup>11</sup>

Continuously compounded returns,  $\dot{r}_t$ , can be converted to the standard unit normal variable

$$z_t = \frac{\dot{r}_t - \dot{\mu}}{\dot{\sigma}} \quad (10)$$

where  $z_t$  is normally distributed with mean 0 and variance 1,  $N(0, 1)$ . This conversion simplifies the derivation of the efficient frontier because the probability distribution of  $z_t$  is stable and the statistical relationships are well known. These conventions imply that  $\dot{r}_t = \ln(1 + r_t) = \dot{\mu} + z_t \dot{\sigma}$ , and that beginning wealth  $W_0$  is related to terminal wealth  $W_T$  as shown in equation (11).

$$W_T = W_0 (1 + r) = W_0 e^{\dot{r}} = W_0 e^{\dot{\mu} + z\dot{\sigma}} \quad (11)$$

Here the subscripts to  $r$  and  $z$  have been deleted for ease of exposition.

To maximize expected utility of terminal wealth, the investor will choose the portfolio that will maximize the expected utility equation (12),

$$E[U(W_T)] = \int_{-\infty}^{+\infty} U[W_0 e^{\mu + z\sigma}] f(z) dz \quad (12)$$

because  $z = g(\mu, \sigma)$ , the portfolio owner's expected utility is a function of expected return  $\mu$  and risk  $\sigma$ , measured using continuously compounded returns. The expected utility function, equation (12), is analyzed next to determine how risk and return affect expected utility.

The partial derivative of expected utility with respect to  $\mu$  will show the effect of maximizing return  $\mu$  for a given risk class  $\sigma$ . Using the chain rule to take the partial derivative of equation (12) yields

$$\begin{aligned} \frac{\partial E[U(W_T)]}{\partial \mu} &= \left( \frac{\partial E[U(W_T)]}{\partial W_T} \right) \left( \frac{\partial W_T}{\partial \mu} \right) \\ &= \int_{-\infty}^{+\infty} \left( \frac{\partial U(W_T)}{\partial W_T} \right) W_T f(z) dz > 0 \end{aligned} \quad (13)$$

because  $(\partial W_T / \partial \mu) = W_T$ . Equation (13) is positive because  $\partial U(W_T) / \partial W_T$  is positive for investors who prefer more to less wealth, and  $f(z)$ , being a probability density function, can only take on positive values. Thus, for any given value of in  $\sigma$  Figure 3B, the investor will seek to maximize  $\mu$  in order to maximize expected utility. Graphically, Figure 3B shows that portfolios on the upper boundary of the ellipse between  $e$  and  $c$  dominate all other portfolios.

The partial derivative of expected utility with respect to  $\sigma$  will show the effect of minimizing  $\sigma$  for a given  $\mu$ . Again, using the chain rule to take the partial derivative of equation (12) yields

$$\begin{aligned} \frac{\partial E[U(W_T)]}{\partial \sigma} &= \left( \frac{\partial E[U(W_T)]}{\partial W_T} \right) \left( \frac{\partial W_T}{\partial \sigma} \right) \\ &= \int_{-\infty}^{+\infty} \left( \frac{\partial U(W_T)}{\partial W_T} \right) W_T z f(z) dz \end{aligned} \quad (14)$$

Equation (14) is identical to equation (13) except the term  $z$  appears in the integrand. Because  $z$  is symmetrically distributed around zero and can therefore take on negative or positive values, and equation (13) was shown to be positive, equation (14) can be either negative or positive (or zero). This means that for a given level of  $\mu$ , expected utility is maximized by either maximizing or minimizing  $\sigma$ . That is,

$$\frac{\partial E[U(W_T)]}{\partial \sigma} = \begin{cases} \text{positive} & \text{if } \left[ \frac{\partial U(W_T)}{\partial W_T} \right] W_T \text{ is an increasing function} \\ 0 & \text{if } \left[ \frac{\partial U(W_T)}{\partial W_T} \right] W_T \text{ is a constant} \\ \text{negative} & \text{if } \left[ \frac{\partial U(W_T)}{\partial W_T} \right] W_T \text{ is a decreasing function} \end{cases} \quad (15)$$

The sign of  $\partial E[U(W_T)]/\partial \dot{\sigma}$  will depend on the sign of the derivative of  $[\partial U(W_T)/\partial W_T] W_T$ , which equals

$$\begin{aligned}\frac{\partial [U'(W_T) \cdot W_T]}{\partial W_T} &= U''(W_T) \cdot W_T + U'(W_T) = U'(W_T) \left[ -\frac{U''(W_T) W_T}{U'(W_T)} + 1 \right] \\ &= U'(W_T) [1 + R(W_T)]\end{aligned}\quad (16)$$

where  $R(W_T) = W_T[-U''(W_T)/U'(W_T)]$  is the utility function's measure of relative risk aversion (RRA). We know

$$\frac{\partial E[U(W_T)]}{\partial \dot{\sigma}} = \begin{cases} \text{positive} & \text{if } R(W_T) > 1 \\ 0 & \text{if } R(W_T) = 1 \\ \text{negative} & \text{if } R(W_T) < 1 \end{cases} \quad (17)$$

Thus, the sign of equation (17) will depend on the investor's wealth and utility function.<sup>12</sup> This means that, without additional information on  $W_T$  and function  $U(\cdot)$ , it is not possible to rule out any part of the upper boundary of the ellipse between  $e$  and  $c$  as being dominated by any other part.

The portfolios on the boundary of the ellipse between  $e$  and  $c$  must correspond to certain portfolios in  $(\mu, \sigma)$  space. Once this correspondence is found, the efficient frontier when returns are lognormal can be identified. Equations (7) and (8) are the key to unraveling these relationships. Consider the ratio of equation (8) divided by equation (7).

$$\frac{\sigma}{\mu} = [\exp(\dot{\sigma}^2) - 1]^{1/2} \quad (18)$$

From equation (18), it can be seen that

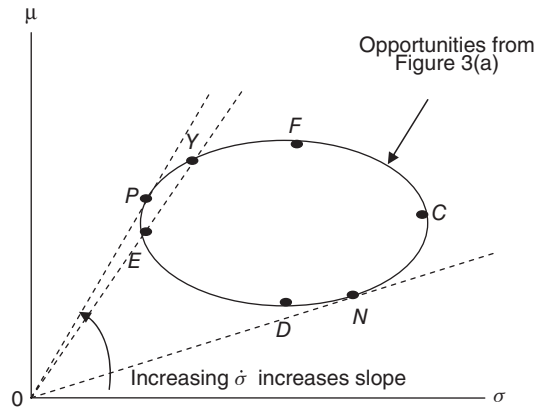
$$\mu = \frac{\sigma}{[\exp(\dot{\sigma}^2) - 1]^{1/2}} \quad (19)$$

Equation (19) represents a straight line from the origin in  $(\mu, \sigma)$  space that has a constant value of  $\dot{\sigma}^2$  throughout its length. To further analyze the investment opportunities in  $(\mu, \sigma)$  space from Figure 3A, a family of such constant  $\dot{\sigma}^2$  lines is illustrated in Figure 4.

Taking the logarithm of equation (9.7) and solving it for  $\dot{\mu}$  yields

$$\dot{\mu} = \ln \mu - 0.5\dot{\sigma}^2 \quad (20)$$

If  $\dot{\sigma}$  is a given value in equation (20), then  $\dot{\mu}$  is made as large as possible by maximizing  $\mu$ . This implies that moving out from the origin along any linear isoquant in Figure 4 to the point on the opportunity set with the highest attainable value of either  $\mu$  or  $\dot{\mu}$  will trace out the unique  $(\mu, \sigma)$  efficient portfolio that corresponds to one  $(\dot{\mu}, \dot{\sigma})$  efficient portfolio. For example, on the ray OEY in Figure 4, portfolio Y has both the maximum  $\dot{\mu}$  and the maximum  $\mu$  along this isoquant. Tracing the maximum  $\dot{\mu}$  portfolios at every value of  $\dot{\sigma}$  in Figure 4 reveals that the curve PYFCN is the  $(\dot{\mu}, \dot{\sigma})$  efficient frontier. The existence of this efficient frontier implies



**FIGURE 4** Analysis of  $(\mu, \sigma)$  Opportunities in Terms of  $\tilde{\mu}$  and  $\tilde{\sigma}$

that  $\tilde{\sigma}^2$  is a quantitative risk surrogate that may be used in place of the original  $\sigma^2$  Markowitz risk measure.

Several interesting observations may be drawn with respect to the  $(\tilde{\mu}, \tilde{\sigma})$  efficient frontier. First, note that efficient portfolio  $f$  in Figure 3B is the portfolio with the maximum expected return in the long run. That is, efficient portfolio  $f$  has the maximum geometric mean rate of return, because  $E[\ln(1+r)] = \tilde{\mu}$  is a geometric mean return.<sup>13</sup> It also corresponds to the optimal portfolio for an investor with a logarithmic utility function and will lie between  $E$  and  $F$  (the traditional efficient frontier) in Figure 3A.<sup>14</sup> Second, note that the portion of the  $(\mu, \sigma)$  efficient frontier along the curve  $EP$  in Figure 4 is not  $(\tilde{\mu}, \tilde{\sigma})$  efficient. That is, a part of the traditional efficient frontier is no longer efficient if returns are lognormal. Third, if a risk-free borrowing and lending rate exists, then there is no difference in the efficient frontiers because the tangency points are identical. Finally, if unrestricted short selling is permitted (but risk-free borrowing and lending are prohibited), then the new efficient frontier will be a subset of the traditional efficient frontier. More specifically, it will be an upper portion of the upper half of the hyperbola, because it will no longer include a portion near the minimum variance portfolio. These last two observations imply that the market portfolio will still lie on the traditional efficient frontier as long as either unrestricted short selling is permitted or a risk-free borrowing and lending rate exists. As a result, under these circumstances the CAPM still holds if returns are lognormally distributed.

Restating the efficient frontier in  $(\tilde{\mu}, \tilde{\sigma})$  space is worthwhile for several reasons. First, the  $(\mu, \sigma)$ -efficient frontier contains some undesirable portfolios in its highest and lowest return sectors that are eliminated in the logarithmic transformation to the  $(\tilde{\mu}, \tilde{\sigma})$ -efficient frontier. Second, the  $(\mu, \sigma)$ -efficient frontier can ignore a significant third moment. The  $(\tilde{\mu}, \tilde{\sigma})$  analysis, however, includes and eliminates the positive skewness of the discrete holding period returns in deriving the continuous-time efficient frontier and is thus a superior decision-making process. Finally, the  $(\tilde{\mu}, \tilde{\sigma})$  analysis pinpoints the long-run wealth-maximizing portfolio, whereas this maximum geometric mean return portfolio is not delineated by the  $(\mu, \sigma)$  analysis. The  $(\tilde{\mu}, \tilde{\sigma})$ -efficient portfolio with the highest expected return is the optimal choice for investors wishing to maximize their terminal wealth.

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## 6 CONCLUSIONS

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The portfolio analysis is based on the assumption that asset returns are normally distributed with two parameters, mean and variance. However, empirical evidence shows that asset returns are not normally distributed. The empirical distribution of asset returns has nonsymmetric and leptokurtic characteristics. In other words, the distribution of asset returns is not characterized by the two parameters. This chapter has reviewed several non-normal probability distributions that are known to better capture the non-normal characteristics of asymmetry and leptokurtosis of asset returns.

## NOTES

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1. For an examination of daily stock return distributions for U.S. stocks, see Rachev, Stoyanov, Biglova, and Fabozzi (2005).
2. The skewness and kurtosis, which are the statistics most commonly used to test normality, are defined and measured by

$$\begin{aligned}\text{Skewness} \equiv \gamma_1 &= E \left[ \left( \frac{r_t - \mu}{\sigma} \right)^3 \right] = \frac{\mu_3}{\sigma^3} \\ &= \frac{\left[ \sum_{t=1}^T (r_t - \bar{r})^3 / T \right]}{\left[ \sum_{t=1}^T (r_t - \bar{r})^2 / T \right]^{3/2}} = \frac{\sqrt{T} \sum_{t=1}^T (r_t - \bar{r})^3}{\left[ \sum_{t=1}^T (r_t - \bar{r})^2 \right]^{3/2}} \\ \text{Kurtosis} \equiv \gamma_2 &= E \left[ \left( \frac{r_t - \mu}{\sigma} \right)^4 \right] - 3 = \frac{\mu_4}{\sigma^4} - 3 \\ &= \frac{\left[ \sum_{t=1}^T (r_t - \bar{r})^4 / T \right]}{\left[ \sum_{t=1}^T (r_t - \bar{r})^2 / T \right]^2} - 3 = \frac{T \sum_{t=1}^T (r_t - \bar{r})^4}{\left[ \sum_{t=1}^T (r_t - \bar{r})^2 \right]^2} - 3\end{aligned}$$

where  $\mu_k = E(r_t - \mu)^k$  is the  $k$ -th central moment,  $\mu = E(r_t)$ , and  $T$  is the sample size of time-series return observations. For normality,  $\gamma_1 = 0$  and  $\gamma_2 = 0$ . The upper and lower one-percentile points in the distribution of the skewness statistic are 0.058 and  $-0.058$ , respectively. The upper and lower one-percentile points in the distribution of the excess kurtosis statistic are 0.13 and  $-0.11$ , respectively (see E. S. Pearson and H. O. Hartley, eds., 1966).

3. Of course, the mean must also be finite. See Fama (1976a, chap. 1) for a more detailed presentation of this counterargument.
4. It has been argued that the characteristic exponent  $\alpha$  lies somewhere between 1.6 and 1.8.
5. The semi-interquartile range is equal to half the difference between the 0.75 and 0.25 fractiles of the cumulative probability distribution. For a summary on this form of two-dimensional portfolio analysis, see Fama and Miller (1972, chaps. 6 and 7).
6. Ortobelli, Rachev, and Fabozzi (2010) analyze portfolio selection problems with stable Paretian distributions.
7. As a practical matter, it can be counterargued that limited liability and positive skewness are unimportant when returns are measured over arbitrarily small intervals.

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8. See Merton (1971) for a set of conditions under which portfolios will have lognormal returns when the component securities have lognormal returns.
  9. For an interesting discussion of Tversky and Kahneman prospect theory and skewness, see Nicholas Barberis and Ming Huang, "Stocks as Lotteries: The Implications of Probability Weighting for Security Prices," *American Economic Review* 98, no. 5: 2066–2100.
  10. See the appendix to Chapter 1 for a description and comparison of discrete and continuously compounded returns.
  11. The use of ellipses here is for illustrative purposes only.
  12. For example, if the investor has a logarithmic utility function, then  $\partial E[U(W_T)] / \partial \dot{\sigma} = 0$ . In this situation, all portfolios with the same level of  $\dot{\mu}$  will be equally desirable, regardless of their respective level of  $\dot{\sigma}$ .
  13. Given that there are  $T$  observed returns on a security, the geometric mean of the security is defined as  $[(1 + r_1)(1 + r_2) \cdots (1 + r_T)]^{1/T} - 1$ . Take logs of the geometric mean and then find the expected value to obtain  $E[\ln(1 + r)]$ .
  14. If an investor has either a quadratic or logarithmic utility function, then his or her optimal portfolio will still be on the traditional efficient frontier even if returns have a lognormal distribution. See Elton and Gruber (1974b).

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